

Clinical Variant Interpretation Report

Variant: chr17:41215920 | Gene: BRCA1 | Class: Pathogenic

1. Variant Summary

- * Variant ID: chr17:41215920
- * Gene: BRCA1
- * Variant Type: Duplication (c.5266dupC)
- * Protein Change: p.Gln1756Profs (premature translational stop signal)
- * Population Frequency: Known founder mutation in the Ashkenazi Jewish population

2. Clinical Classification

- * Classification: Pathogenic
- * Clinical Condition: Hereditary breast and ovarian cancer syndrome
- * Clinical Significance: This variant is associated with an increased risk of developing hereditary breast and ovarian cancer syndrome.

3. Supporting Evidence

- * Functional Assays: Complete loss of homology-directed DNA repair capability has been demonstrated in functional assays.
- * Population Studies: This variant is a known founder mutation in the Ashkenazi Jewish population, indicating a significant association with hereditary breast and ovarian cancer syndrome.
- * Literature Review: Definitive link to hereditary breast and ovarian cancer syndrome has been established in the literature.